

q:-

Brute Force Maximum Subarray

Given an array $[-2, 3, 4, -1, -2, 1, 5, -3]$

$$A = [-2, 3, 4, -1, -2, 1, 5, -3]$$

Steps:-

1. Select a starting index.
2. Index - extend the subarray one element at a time.
3. Calculate the sum of each subarray.
4. compare it with the current maximum sum.

An array size of $n = 8$

$$\begin{aligned}\frac{n(n+1)}{2} &= \frac{8(9)}{2} \\ &= \frac{36}{2} \\ &= 36\end{aligned}$$

Therefore, 36 subarrays are examined.

calculations of Different subarrays

Index 0

subarray	sum
$[-2]$	-2
$[-2, 3]$	1
$[-2, 3, 4]$	5
$[-2, 3, 4, -1]$	4
$[-2, 3, 4, -1, 2]$	2
$[-2, 3, 4, -1, 2, 1]$	3
$[-2, 3, 4, -1, -2, 1, 5]$	8
$[-2, 3, 4, -1, -2, 1, 5, -3]$	5

Index 1

subarray	sum
$[3]$	3
$[3, 4]$	7
$[3, 4, -1]$	6
$[3, 4, -1, -2]$	4

[3, 4, -1, -2, 1]

5

[3, 4, -1, -2, 1, 5]

10

[3, 4, -1, -2, 1, 5, -3]

7

Index 2

Subarray

Sum

[4]

4

[4, -1]

3

[4, -1, -2]

1

[4, -1, -2, 1]

2

[4, -1, -2, 1, 5]

7

[4, -1, -2, 1, 5, -3]

4

Index 3

Subarray

Sum

[-1]

-1

[-1, -2]

-3

[-1, -2, 1]

-2

[-1, -2, 1, 5]

3

[-1, -2, 1, 5, -3]

0

Index 4.

Subarray

Sum

$[-2]$

-2

$[-2, 1]$

-1

$[-2, 1, 5]$

4

$[-2, 1, 5, -3]$

1

Index 5

Subarray

Sum

$[1]$

1

$[1, 5]$

6

$[1, 5, -3]$

3

Index 6

Subarray

Sum

$[5]$

5

$[5, -3]$

2

Index 7

Subarray

Sum

$[-3]$

-3

Number of comparisons:-

36 subarrays is compared with the current maximum sum.

Maximum sum:-

The highest sum obtained is

$$\boxed{10}$$

$$[3, 4, -1, -2, 1, 5]$$

Verification

$$3 + 4 - 1 - 2 + 1 + 5 = 10$$

Time complexity:-

$$T(n) = O(n^2)$$

$$O(n^3)$$

Space complexity:-

(sum & Max sum)

$$\boxed{O(1)}$$